

# Analysis of Linear Programming

## Part 3b: The Simplex Method

IE 311 – Operations Research I

## Recall: Linear Programming (Standard Form)

Any Linear Program can be written in **standard form** – only equality constraints and nonnegative variables:

$$\min; c^T x \quad \text{s.t.} \quad Ax = b, \quad x \geq 0.$$

The simplex method operates on this form, moving from one basic feasible solution (extreme point) to an adjacent one while improving the objective.

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 Method

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Method

Algorithm used to solve continuous models

|                |     |
|----------------|-----|
| Type:          | int |
| Default value: | -1  |
| Minimum value: | -1  |
| Maximum value: | 5   |

Algorithm used to solve continuous models or the root node of a MIP model. Options are: -1=automatic, [0=primal simplex](#), [1=dual simplex](#), 2=barrier, 3=concurrent, 4=deterministic concurrent, 5=deterministic concurrent simplex.

# Main Steps of the (Primal) Simplex Method

The simplex algorithm proceeds as follows:

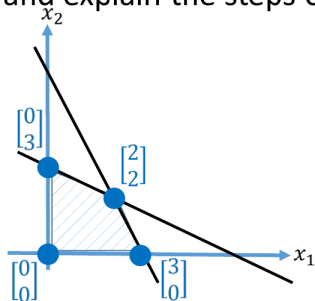
- **Step 1:** Convert the LP to standard form.
- **Step 2:** Obtain a basic feasible solution (BFS), if possible, from the standard form.
- **Step 3:** Determine whether the current BFS is optimal.
- **Step 4:** If the current BFS is not optimal, determine which nonbasic variable should become basic and which basic variable should become nonbasic, to obtain a new BFS with a better objective value (or detect unboundedness).
- **Step 5:** Use elementary row operations (EROs) to obtain that new BFS, then go back to Step 3.

- Although the simplex method can solve any LP in principle, we first look at **structured** LPs (those with an obvious initial BFS) before turning to **unstructured** LPs (those without one).
- In particular, we first learn to solve **maximization problems with only  $\leq$  constraints and a nonnegative right-hand side  $b$**  (an obvious BFS is then available).
- Then we discuss how to solve **minimization problems** with minor modifications.
- Finally, we learn how to incorporate  $\geq$  and  $=$  constraints with an arbitrary right-hand side (these require preliminary work – the Big-M or Two-Phase methods – to obtain an initial BFS).

## A First Example

- We will first go over the following example LP and explain the steps of the Simplex Method.

$$\begin{array}{ll}\max & x_0 = 3x_1 + 4x_2 \\ \text{s.t.} & 2x_1 + x_2 \leq 6 \\ & x_1 + 2x_2 \leq 6 \\ & x_1, x_2 \geq 0.\end{array}$$



- For convenience, we defined a new variable  $x_0$  (**column-0**), which represents the objective function value (**row-0**).

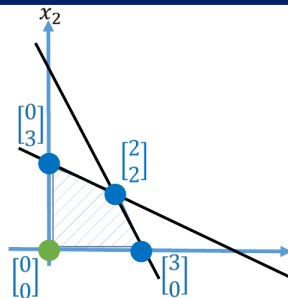
- In Step 1, we convert the LP to standard form:

$$\begin{array}{ll}\max & x_0 = 3x_1 + 4x_2 \\ \text{s.t.} & 2x_1 + x_2 + x_3 = 6 \\ & x_1 + 2x_2 + x_4 = 6 \\ & x_1, x_2, x_3, x_4 \geq 0.\end{array}$$

## A First Example: Step 2

- In Step 2, we obtain an initial BFS:

| Row | BV    |       |         |         |        | RHS   |
|-----|-------|-------|---------|---------|--------|-------|
| 0   | $x_0$ | $x_0$ | $-3x_1$ | $-4x_2$ |        | $= 0$ |
| 1   | $x_3$ |       | $2x_1$  | $+x_2$  | $+x_3$ | $= 6$ |
| 2   | $x_4$ |       | $x_1$   | $+2x_2$ | $+x_4$ | $= 6$ |



- Convention:** Variable  $x_0$  will ALWAYS be basic.
- Note:** Since the original LP has only  $\leq$  constraints with nonnegative RHS, the slack variables  $x_3$  and  $x_4$  are chosen as basic variables.
- RHS column** in this tableau gives the current values of the basic variables.

## A First Example: Step 3

- In Step 3, we check whether the current BFS is optimal (**look at row-0**):

| Row | BV    |       |         |         |        | RHS   |
|-----|-------|-------|---------|---------|--------|-------|
| 0   | $x_0$ | $x_0$ | $-3x_1$ | $-4x_2$ |        | $= 0$ |
| 1   | $x_3$ |       | $2x_1$  | $+x_2$  | $+x_3$ | $= 6$ |
| 2   | $x_4$ |       | $x_1$   | $+2x_2$ | $+x_4$ | $= 6$ |

- Idea:** If we move to an adjacent BFS, can we improve the objective function variable  $x_0$ ?
- Moving to an adjacent BFS means that we make one of the nonbasic variables basic while making one of the basic variables nonbasic.
- Note that increasing the value of  $x_1$  or  $x_2$  would make  $x_0$  larger (WHY?)**
- Therefore, the current BFS is NOT proven to be optimal.

- In Step 4, we first decide which nonbasic variable to make basic (**look at row-0**):

| Row | BV    |       |         |         |        | RHS        |
|-----|-------|-------|---------|---------|--------|------------|
| 0   | $x_0$ | $x_0$ | $-3x_1$ | $-4x_2$ |        | $= 0$      |
| 1   | $x_3$ |       | $2x_1$  | $+x_2$  | $+x_3$ | $= 6$      |
| 2   | $x_4$ |       | $x_1$   | $+2x_2$ |        | $+x_4 = 6$ |

- Idea:** Intuitively, choosing the most negative coefficient in row-0 might improve the objective largest (although any nonbasic variable with negative coefficient can be chosen).
- In this case, we can choose the nonbasic variable  $x_2$  and make it basic.**
- Such a variable will be called *the entering variable*.

- In Step 4, we next decide which basic variable to make nonbasic (**look at the column of the entering variable and RHS**):

| Row | BV    |       |         |         | RHS    |       |
|-----|-------|-------|---------|---------|--------|-------|
| 0   | $x_0$ | $x_0$ | $-3x_1$ | $-4x_2$ | $= 0$  |       |
| 1   | $x_3$ |       | $2x_1$  | $+x_2$  | $+x_3$ | $= 6$ |
| 2   | $x_4$ |       | $x_1$   | $+2x_2$ | $+x_4$ | $= 6$ |

- Idea:** What is the adjacent BFS obtained by making  $x_2$  basic?
- We need to perform the so-called **minimum ratio test**:
  - Obtain the entrywise ratio of the RHS and the column of the entering variable among the rows that have positive entry in the column of the entering variable.
  - Choose the basic variable that gives the minimum ratio as *the leaving variable* (if there is no such variable, the LP is unbounded):

The leaving variable is  $x_3$  since  $\min \left\{ \frac{6}{1}, \frac{6}{2} \right\} = 3$

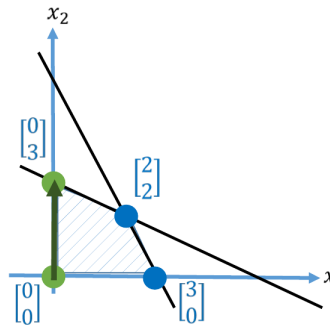
## A First Example: Step 5

- In Step 5, we use EROs to find a new BFS with a better objective function value (**pivoting**):

| Row | BV    |       |         |         |        | RHS   |
|-----|-------|-------|---------|---------|--------|-------|
| 0   | $x_0$ | $x_0$ | $-3x_1$ | $-4x_2$ |        | $= 0$ |
| 1   | $x_3$ |       | $2x_1$  | $+x_2$  | $+x_3$ | $= 6$ |
| 2   | $x_4$ |       | $x_1$   | $+2x_2$ | $+x_4$ | $= 6$ |

- Idea:** How can we make the column of the entering variable a unit vector by pivoting?

| pivoting? |       |       |                  |        |                         | RHS    |
|-----------|-------|-------|------------------|--------|-------------------------|--------|
| 0         | $x_0$ | $x_0$ | $-x_1$           |        | $+2x_4$                 | $= 12$ |
| 1         | $x_3$ |       | $\frac{3}{2}x_1$ |        | $+x_3 - \frac{1}{2}x_4$ | $= 3$  |
| 2         | $x_2$ |       | $\frac{1}{2}x_1$ | $+x_2$ | $+\frac{1}{2}x_4$       | $= 3$  |



## A First Example: Step 3 (Again)

- In Step 3, we check whether the current BFS is optimal (**look at row-0**):

| Row | BV    |       |                  |        |                   | RHS    |
|-----|-------|-------|------------------|--------|-------------------|--------|
| 0   | $x_0$ | $x_0$ | $-x_1$           |        | $+2x_4$           | $= 12$ |
| 1   | $x_3$ |       | $\frac{3}{2}x_1$ | $+x_3$ | $-\frac{1}{2}x_4$ | $= 3$  |
| 2   | $x_2$ |       | $\frac{1}{2}x_1$ | $+x_2$ | $+\frac{1}{2}x_4$ | $= 3$  |

- Idea:** If we move to an adjacent BFS, can we improve the objective function variable  $x_0$ ?
- Note that increasing the value of  $x_1$  would make  $x_0$  larger.**
- Therefore, the current BFS is NOT proven to be optimal.

## A First Example: Step 4 (Again)

- In Step 4, we first decide which nonbasic variable to make basic (**look at row-0**):

| Row | BV    |                  |        |                   | RHS    |
|-----|-------|------------------|--------|-------------------|--------|
| 0   | $x_0$ | $x_0$            | $-x_1$ | $+2x_4$           | $= 12$ |
| 1   | $x_3$ | $\frac{3}{2}x_1$ | $+x_3$ | $-\frac{1}{2}x_4$ | $= 3$  |
| 2   | $x_2$ | $\frac{1}{2}x_1$ | $+x_2$ | $+\frac{1}{2}x_4$ | $= 3$  |

- Idea:** Intuitively, choosing the most negative coefficient in row-0 might improve the objective largest (although any nonbasic variable with negative coefficient can be chosen).
- In this case, there is only one candidate: the nonbasic variable  $x_1$  enters the basis.**

In Step 4, we next decide which basic variable to make nonbasic (**look at the column of the entering variable and RHS**):

| Row | BV    |       |                  |        |                         | RHS                    |
|-----|-------|-------|------------------|--------|-------------------------|------------------------|
| 0   | $x_0$ | $x_0$ | $-x_1$           |        | $+2x_4$                 | $= 12$                 |
| 1   | $x_3$ |       | $\frac{3}{2}x_1$ |        | $+x_3 - \frac{1}{2}x_4$ | $= \textcolor{red}{3}$ |
| 2   | $x_2$ |       | $\frac{1}{2}x_1$ | $+x_2$ | $+\frac{1}{2}x_4$       | $= \textcolor{red}{3}$ |

- **Idea:** What is the adjacent BFS obtained by making  $x_2$  basic?
- We perform the **minimum ratio test**:

The leaving variable is  $x_3$  since  $\min \left\{ \frac{3}{3/2}, \frac{3}{1/2} \right\} = 2$ .

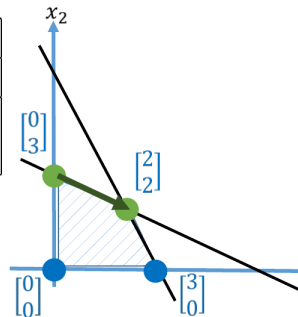
## A First Example: Step 5 (Again)

- In Step 5, we use EROs to find a new BFS with a better objective function value (**pivoting**):

| Row | BV    |       |                  |        |                   | RHS    |
|-----|-------|-------|------------------|--------|-------------------|--------|
| 0   | $x_0$ | $x_0$ | $-x_1$           |        | $+2x_4$           | $= 12$ |
| 1   | $x_3$ |       | $\frac{3}{2}x_1$ | $+x_3$ | $-\frac{1}{2}x_4$ | $= 3$  |
| 2   | $x_2$ |       | $\frac{1}{2}x_1$ | $+x_2$ | $+\frac{1}{2}x_4$ | $= 3$  |

- Idea:** How can we make the column of the entering variable a unit vector by pivoting?

| Row | BV    |       |       |                   |                   | RHS    |
|-----|-------|-------|-------|-------------------|-------------------|--------|
| 0   | $x_0$ | $x_0$ |       | $+\frac{2}{3}x_3$ | $+\frac{5}{3}x_4$ | $= 14$ |
| 1   | $x_1$ |       | $x_1$ | $+\frac{2}{3}x_3$ | $-\frac{1}{3}x_4$ | $= 2$  |
| 2   | $x_2$ |       | $x_2$ | $-\frac{1}{3}x_3$ | $+\frac{2}{3}x_4$ | $= 2$  |



## A First Example: Step 3 (Again)

- In Step 3, we check whether the current BFS is optimal (**look at row-0**):

| Row | BV    |       |                                     | RHS    |
|-----|-------|-------|-------------------------------------|--------|
| 0   | $x_0$ | $x_0$ | $+\frac{2}{3}x_3$ $+\frac{5}{3}x_4$ | $= 14$ |
| 1   | $x_1$ | $x_1$ | $+\frac{2}{3}x_3$ $-\frac{1}{3}x_4$ | $= 2$  |
| 2   | $x_2$ | $x_2$ | $-\frac{1}{3}x_3$ $+\frac{2}{3}x_4$ | $= 2$  |

- Idea:** If we move to an adjacent BFS, can we improve the objective function variable  $x_0$ ?
- Note that increasing the value of  $x_3$  or  $x_4$  would make  $x_0$  smaller.**
- Therefore, the current BFS is **OPTIMAL** with objective value  $x_0^* = 14$ .
- Optimal solution:  $x_1^* = x_2^* = 2$ .

- Consider the following *structured* LP with  $b \geq 0$ :

$$\begin{aligned} \max \quad & \sum_{j=1}^n c_j x_j \\ \text{s.t.} \quad & \sum_{j=1}^n a_{ij} x_j \leq b_i \quad i = 1, \dots, m \\ & x_j \geq 0 \quad j = 1, \dots, n. \end{aligned}$$

- **Step 1** Convert the LP to standard form.

$$\max x_0 = \sum_{j=1}^n c_j x_j$$

$$\text{s.t.} \quad \sum_{j=1}^n a_{ij} x_j + x_{n+i} = b_i \quad i = 1, \dots, m$$

$$x_j \geq 0 \quad j = 1, \dots, n+m.$$

- **Step 2** Obtain a BFS from the standard form.
- Note that the slack variables  $x_{n+1}, \dots, x_{n+m}$  constitute an *obvious* basis (since  $b \geq 0$ ).
- We construct the following *simplex tableau*:

| Row      | BV        |       |             |         |              |            | RHS      |
|----------|-----------|-------|-------------|---------|--------------|------------|----------|
| 0        | $x_0$     | $x_0$ | $-c_1x_1$   | $\dots$ | $-c_nx_n$    |            | $= 0$    |
| 1        | $x_{n+1}$ |       | $a_{11}x_1$ | $\dots$ | $+a_{1n}x_n$ | $+x_{n+1}$ | $= b_1$  |
| $\vdots$ | $\vdots$  |       | $\dots$     | $\dots$ | $\vdots$     | $\dots$    | $\vdots$ |
| $m$      | $x_{n+m}$ |       | $a_{m1}x_1$ | $\dots$ | $+a_{mn}x_n$ | $+x_{n+m}$ | $= b_m$  |

### **Step 3 – Determine whether the current BFS is optimal.**

Look at the coefficients of row 0 (the objective row) of the simplex tableau, except  $z_0$ .

- If **all** these entries are nonnegative, then STOP: the current BFS is optimal.
- Otherwise, we cannot prove optimality; proceed to Step 4.

### **Step 4a – Determine which nonbasic variable should become basic.**

Any nonbasic variable with a **negative** coefficient in row 0 may be chosen as the entering variable (this guarantees the objective value will improve, or at least not deteriorate). As a common practice, we choose the variable with the **most negative** coefficient.

### Step 4b – Determine which basic variable should become nonbasic.

Let  $k$  be the entering variable,  $a_{ik}$  the entries of its column, and  $b_i$  the right-hand side in the current tableau (ignoring row 0). Apply the **minimum ratio test**:

$$i^* \in \arg \min_{i: a_{ik} > 0} \frac{b_i}{a_{ik}}.$$

- If no row has  $a_{ik} > 0$ , then STOP: the LP is **unbounded**.
- Otherwise, the basic variable in row  $i^*$  leaves the basis (break ties by smallest index).

### **Step 5 – Obtain the new BFS.**

Use elementary row operations (EROs) to pivot on the entry  $a_{i^*k}$ : the entering variable becomes basic and the leaving variable becomes nonbasic. This yields a new BFS with a better objective value. Then return to Step 3.

- The value of the entering variable in the next BFS equals the **minimum ratio**.
- The change in the objective value equals the minimum ratio times the negative of the entering variable's row-0 coefficient.
- If there is a **tie** in the minimum ratio, the next BFS is **degenerate**.
- If a nonbasic variable has a row-0 coefficient of **0** in the optimal tableau, the LP may have **multiple (alternative) optima**.
- If the LP is **unbounded**, the entering variable's column gives a direction of unboundedness that makes a positive inner product with the objective.

- Solve the following LP using the simplex method:

$$\max x_0 = 6x_1 + 3x_2 + 2x_3$$

$$\text{s.t.} \quad 8x_1 + 6x_2 + x_3 \leq 48$$

$$4x_1 + 2x_2 + \frac{3}{2}x_3 \leq 20$$

$$2x_1 + \frac{3}{2}x_2 + \frac{1}{2}x_3 \leq 8$$

- Before pivoting, state the value of the entering variable in the next BFS and the change in the objective function value.

## Example 2: Iteration 1

| Row | BV    |       |         |                   |                   |        | RHS    |
|-----|-------|-------|---------|-------------------|-------------------|--------|--------|
| 0   | $x_0$ | $x_0$ | $-6x_1$ | $-3x_2$           | $-2x_3$           |        | $= 0$  |
| 1   | $x_4$ |       | $8x_1$  | $+6x_2$           | $+x_3$            | $+x_4$ | $= 48$ |
| 2   | $x_5$ |       | $4x_1$  | $+2x_2$           | $+\frac{3}{2}x_3$ | $+x_5$ | $= 20$ |
| 3   | $x_6$ |       | $2x_1$  | $+\frac{3}{2}x_2$ | $+\frac{1}{2}x_3$ | $+x_6$ | $= 8$  |

- **Entering Variable:**  $x_1$  (most negative coefficient =  $-6$ )
- **Leaving Variable:**  $x_6$  (minimum ratio = 4)

| Row | BV    |       |                   |                   |                   |                   | RHS    |
|-----|-------|-------|-------------------|-------------------|-------------------|-------------------|--------|
| 0   | $x_0$ | $x_0$ |                   | $+\frac{3}{2}x_2$ | $-\frac{1}{2}x_3$ | $+3x_6$           | $= 24$ |
| 1   | $x_4$ |       |                   |                   | $-x_3$            | $+x_4$            | $= 16$ |
| 2   | $x_5$ |       |                   | $-x_2$            | $+\frac{1}{2}x_3$ | $+x_5$            | $= 4$  |
| 3   | $x_1$ | $x_1$ | $+\frac{3}{4}x_2$ | $+\frac{1}{4}x_3$ |                   | $+\frac{1}{2}x_6$ | $= 4$  |

## Example 2: Iteration 2

| Row | BV    |       |                   |                   |        |                   | RHS    |
|-----|-------|-------|-------------------|-------------------|--------|-------------------|--------|
| 0   | $x_0$ | $x_0$ | $+\frac{3}{2}x_2$ | $-\frac{1}{2}x_3$ |        | $+3x_6$           | $= 24$ |
| 1   | $x_4$ |       |                   | $-x_3$            | $+x_4$ | $-4x_6$           | $= 16$ |
| 2   | $x_5$ |       | $-x_2$            | $+\frac{1}{2}x_3$ | $+x_5$ | $-2x_6$           | $= 4$  |
| 3   | $x_1$ | $x_1$ | $+\frac{3}{4}x_2$ | $+\frac{1}{4}x_3$ |        | $+\frac{1}{2}x_6$ | $= 4$  |

- **Entering Variable:**  $x_3$  (most negative coefficient =  $-1/2$ )
- **Leaving Variable:**  $x_5$  (minimum ratio = 8)

| Row | BV    |       |                   |        |                   |                   | RHS    |
|-----|-------|-------|-------------------|--------|-------------------|-------------------|--------|
| 0   | $x_0$ | $x_0$ | $+\frac{1}{2}x_2$ |        | $+x_5$            | $+x_6$            | $= 28$ |
| 1   | $x_4$ |       | $-2x_2$           | $+x_4$ | $+2x_5$           | $-8x_6$           | $= 24$ |
| 2   | $x_3$ |       | $-2x_2$           | $+x_3$ | $+2x_5$           | $-4x_6$           | $= 8$  |
| 3   | $x_1$ | $x_1$ | $+\frac{5}{4}x_2$ |        | $-\frac{1}{2}x_5$ | $+\frac{3}{2}x_6$ | $= 2$  |

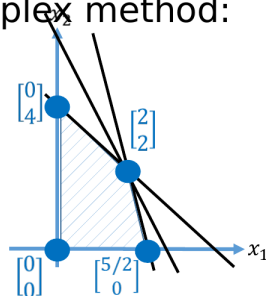
## Example 2: Iteration 3

| Row | BV    |       |                   |        |                   |                   | RHS    |
|-----|-------|-------|-------------------|--------|-------------------|-------------------|--------|
| 0   | $x_0$ | $x_0$ | $+\frac{1}{2}x_2$ |        | $+x_5$            | $+x_6$            | $= 28$ |
| 1   | $x_4$ |       | $-2x_2$           | $+x_4$ | $+2x_5$           | $-8x_6$           | $= 24$ |
| 2   | $x_3$ |       | $-2x_2$           | $+x_3$ | $+2x_5$           | $-4x_6$           | $= 8$  |
| 3   | $x_1$ | $x_1$ | $+\frac{5}{4}x_2$ |        | $-\frac{1}{2}x_5$ | $+\frac{3}{2}x_6$ | $= 2$  |

- **STOP:** This is the optimal tableau (there is a unique optimal solution since objective coefficients of all nonbasic variable are positive).
- Optimal value:  $x_0^* = 28$ .
- Optimal solution:  $x_1^* = 2, x_2^* = 0, x_3^* = 8$ .

- If there is a **tie in the minimum ratio**, the next BFS is degenerate.
- Solve the following LP using the simplex method:

$$\begin{aligned}
 \max \quad & x_0 = 5x_1 + 3x_2 \\
 \text{s.t.} \quad & 4x_1 + 2x_2 \leq 12 \\
 & 4x_1 + x_2 \leq 10 \\
 & x_1 + x_2 \leq 4 \\
 & x_1, x_2 \geq 0.
 \end{aligned}$$

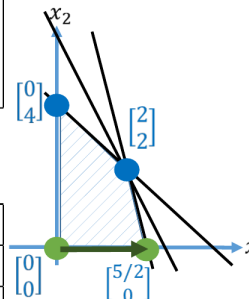


### Example 3: Iteration 1

| Row | BV    |       |         |         |        | RHS        |
|-----|-------|-------|---------|---------|--------|------------|
| 0   | $x_0$ | $x_0$ | $-5x_1$ | $-3x_2$ |        | $= 0$      |
| 1   | $x_3$ |       | $4x_1$  | $+2x_2$ | $+x_3$ | $= 12$     |
| 2   | $x_4$ |       | $4x_1$  | $+x_2$  | $+x_4$ | $= 10$     |
| 3   | $x_5$ |       | $x_1$   | $+x_2$  |        | $+x_5 = 4$ |

- **Entering Variable:**  $x_1$  (most negative coefficient =  $-5$ )
- **Leaving Variable:**  $x_4$  (minimum ratio =  $5/2$ )

| Row | BV    |       |                   |                   |                  | RHS             |
|-----|-------|-------|-------------------|-------------------|------------------|-----------------|
| 0   | $x_0$ | $x_0$ | $-\frac{7}{4}x_2$ | $+\frac{5}{4}x_4$ | $= \frac{25}{2}$ |                 |
| 1   | $x_3$ |       | $x_2$             | $+x_3$            | $-x_4$           | $= 2$           |
| 2   | $x_1$ | $x_1$ | $+\frac{1}{4}x_2$ | $+\frac{1}{4}x_4$ |                  | $= \frac{5}{2}$ |
| 3   | $x_5$ |       | $+\frac{3}{4}x_2$ | $-\frac{1}{4}x_4$ | $+x_5$           | $= \frac{3}{2}$ |

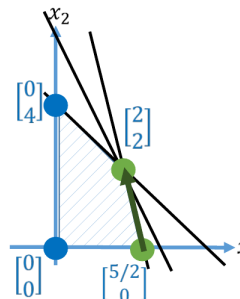


### Example 3: Iteration 2

| Row | BV    |       |                   |                   |        | RHS              |
|-----|-------|-------|-------------------|-------------------|--------|------------------|
| 0   | $x_0$ | $x_0$ | $-\frac{7}{4}x_2$ | $+\frac{5}{4}x_4$ |        | $= \frac{25}{2}$ |
| 1   | $x_3$ |       | $x_2$             | $+x_3$            | $-x_4$ | $= 2$            |
| 2   | $x_1$ | $x_1$ | $+\frac{1}{4}x_2$ | $+\frac{1}{4}x_4$ |        | $= \frac{5}{2}$  |
| 3   | $x_5$ |       | $+\frac{3}{4}x_2$ | $-\frac{1}{4}x_4$ | $+x_5$ | $= \frac{3}{2}$  |

- **Entering Variable:**  $x_2$  (most negative coefficient =  $-7/4$ )
- **Leaving Variable:**  $x_3$  (minimum ratio = 2 – **there is a tie!**)

| Row | BV    |       |                   |                   |        | RHS    |
|-----|-------|-------|-------------------|-------------------|--------|--------|
| 0   | $x_0$ | $x_0$ | $+\frac{7}{4}x_3$ | $-\frac{1}{2}x_4$ |        | $= 16$ |
| 1   | $x_2$ |       | $x_2$             | $+x_3$            | $-x_4$ | $= 2$  |
| 2   | $x_1$ | $x_1$ | $-\frac{1}{4}x_3$ | $+\frac{1}{2}x_4$ |        | $= 2$  |
| 3   | $x_5$ |       | $-\frac{3}{4}x_3$ | $+\frac{1}{2}x_4$ | $+x_5$ | $= 0$  |

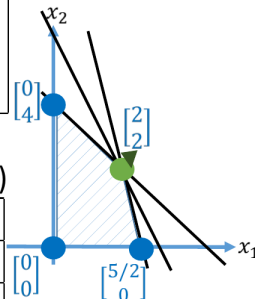


### Example 3: Iteration 3

| Row | BV    |       |                   |                   | RHS        |
|-----|-------|-------|-------------------|-------------------|------------|
| 0   | $x_0$ | $x_0$ | $+\frac{7}{4}x_3$ | $-\frac{1}{2}x_4$ | $= 16$     |
| 1   | $x_2$ | $x_2$ | $+x_3$            | $-x_4$            | $= 2$      |
| 2   | $x_1$ | $x_1$ | $-\frac{1}{4}x_3$ | $+\frac{1}{2}x_4$ | $= 2$      |
| 3   | $x_5$ |       | $-\frac{3}{4}x_3$ | $+\frac{1}{2}x_4$ | $+x_5 = 0$ |

- **Entering Variable:**  $x_4$  (most negative coefficient  $= -1/2$ )
- **Leaving Variable:**  $x_5$  (minimum ratio  $= 0$  – **degenerate pivot**)

| Row | BV    |       |                   |         | RHS         |
|-----|-------|-------|-------------------|---------|-------------|
| 0   | $x_0$ | $x_0$ | $+x_3$            | $+x_5$  | $= 16$      |
| 1   | $x_2$ | $x_2$ | $-\frac{1}{2}x_3$ | $+2x_5$ | $= 2$       |
| 2   | $x_1$ | $x_1$ | $+\frac{1}{2}x_3$ | $-x_5$  | $= 2$       |
| 3   | $x_4$ |       | $-\frac{3}{2}x_3$ | $+x_4$  | $+2x_5 = 0$ |



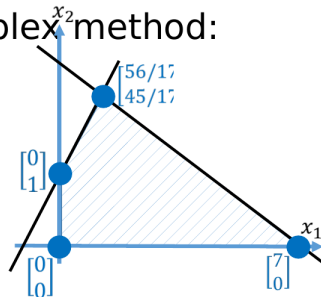
### Example 3: Iteration 4

| Row | BV    |       |                   |         |         | RHS    |
|-----|-------|-------|-------------------|---------|---------|--------|
| 0   | $x_0$ | $x_0$ | $+x_3$            | $+x_5$  |         | $= 16$ |
| 1   | $x_2$ | $x_2$ | $-\frac{1}{2}x_3$ | $+2x_5$ |         | $= 2$  |
| 2   | $x_1$ | $x_1$ | $+\frac{1}{2}x_3$ | $-x_5$  |         | $= 2$  |
| 3   | $x_4$ |       | $-\frac{3}{2}x_3$ | $+x_4$  | $+2x_5$ | $= 0$  |

- **STOP:** This is the optimal tableau (there is a unique optimal solution since objective coefficients of all nonbasic variable are positive).
- Optimal value:  $x_0^* = 16$ .
- Optimal solution:  $x_1^* = x_2^* = 2$ .

- If the objective coefficient of a nonbasic variable is 0 in the optimal tableau, then there exist **multiple (or alternative) optima**.
- Solve the following LP using the simplex method:

$$\begin{array}{ll}\max x_0 = & -3x_1 + 6x_2 \\ \text{s.t.} & 5x_1 + 7x_2 \leq 35 \\ & -x_1 + 2x_2 \leq 2 \\ & x_1, x_2 \geq 0.\end{array}$$

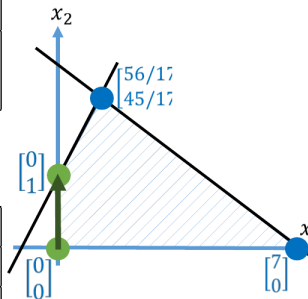


## Example 4: Iteration 1

| Row | BV    |       |         |         |        | RHS    |
|-----|-------|-------|---------|---------|--------|--------|
| 0   | $x_0$ | $x_0$ | $+3x_1$ | $-6x_2$ |        | $= 0$  |
| 1   | $x_3$ |       | $5x_1$  | $+7x_2$ | $+x_3$ | $= 35$ |
| 2   | $x_4$ |       | $-x_1$  | $+2x_2$ |        | $= 2$  |

- **Entering Variable:**  $x_2$  (most negative coefficient =  $-6$ )
- **Leaving Variable:**  $x_4$  (minimum ratio = 1)

| Row | BV    |       |                   |        |                   | RHS    |
|-----|-------|-------|-------------------|--------|-------------------|--------|
| 0   | $x_0$ | $x_0$ | $+0x_1$           |        | $+3x_4$           | $= 6$  |
| 1   | $x_3$ |       | $\frac{17}{2}x_1$ | $+x_3$ | $-\frac{7}{2}x_4$ | $= 28$ |
| 2   | $x_2$ |       | $-\frac{1}{2}x_1$ | $+x_2$ | $+\frac{1}{2}x_4$ | $= 1$  |

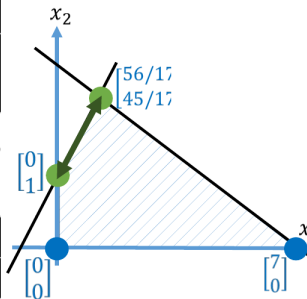


## Example 4: Iteration 2

| Row | BV    |       |                   |        |                   | RHS    |
|-----|-------|-------|-------------------|--------|-------------------|--------|
| 0   | $x_0$ | $x_0$ | $+0x_1$           |        | $+3x_4$           | $= 6$  |
| 1   | $x_3$ |       | $\frac{17}{2}x_1$ | $+x_3$ | $-\frac{7}{2}x_4$ | $= 28$ |
| 2   | $x_2$ |       | $-\frac{1}{2}x_1$ | $+x_2$ | $+\frac{1}{2}x_4$ | $= 1$  |

- **STOP:** This is an optimal tableau (multiple optima exist).
- **What happens if we make  $x_1$  enter the basis?**

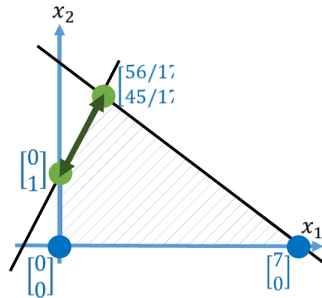
| Row | BV    |       |                    |                    | RHS               |
|-----|-------|-------|--------------------|--------------------|-------------------|
| 0   | $x_0$ | $x_0$ | $+0x_3$            | $+3x_4$            | $= 6$             |
| 1   | $x_1$ | $x_1$ | $+\frac{2}{17}x_3$ | $-\frac{7}{17}x_4$ | $= \frac{56}{17}$ |
| 2   | $x_2$ | $x_2$ | $+\frac{1}{17}x_3$ | $+\frac{5}{17}x_4$ | $= \frac{45}{17}$ |



## Example 4: Follow-up

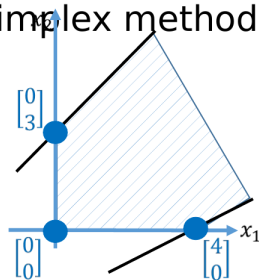
- The set of **all optimal solutions** lie in the following line segment:

$$\left\{ \lambda \begin{bmatrix} 0 \\ 1 \end{bmatrix} + (1 - \lambda) \begin{bmatrix} 56/17 \\ 45/17 \end{bmatrix} : \lambda \in [0, 1] \right\}$$



- If **the LP is unbounded**, the column of the entering variable can be used to determine a direction of unboundedness which makes positive inner product with the objective vector.
- Solve the following LP using the simplex method:

$$\begin{aligned}
 \max \quad & x_0 = x_1 + 3x_2 \\
 \text{s.t.} \quad & x_1 - 2x_2 \leq 4 \\
 & -2x_1 + x_2 \leq 3 \\
 & x_1, x_2 \geq 0.
 \end{aligned}$$

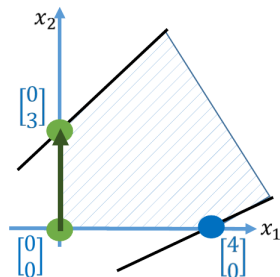


## Example 5: Iteration 1

| Row | BV    |       |         |         |        | RHS        |
|-----|-------|-------|---------|---------|--------|------------|
| 0   | $x_0$ | $x_0$ | $-x_1$  | $-3x_2$ |        | $= 0$      |
| 1   | $x_3$ |       | $x_1$   | $-2x_2$ | $+x_3$ | $= 4$      |
| 2   | $x_4$ |       | $-2x_1$ | $+x_2$  |        | $+x_4 = 3$ |

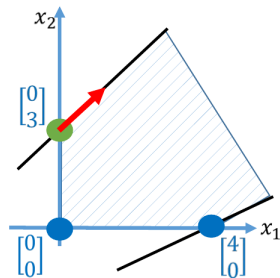
- **Entering Variable:**  $x_2$  (most negative coefficient =  $-3$ )
- **Leaving Variable:**  $x_4$  (minimum ratio = 3)

| Row | BV    |       |         |        |         | RHS    |
|-----|-------|-------|---------|--------|---------|--------|
| 0   | $x_0$ | $x_0$ | $-7x_1$ |        | $+3x_4$ | $= 9$  |
| 1   | $x_3$ |       | $-3x_1$ | $+x_3$ | $+2x_4$ | $= 10$ |
| 2   | $x_2$ |       | $-2x_1$ | $+x_2$ | $+x_4$  | $= 3$  |



## Example 5: Iteration 2

| Row | BV    |       |         |        |         | RHS    |
|-----|-------|-------|---------|--------|---------|--------|
| 0   | $x_0$ | $x_0$ | $-7x_1$ |        | $+3x_4$ | $= 9$  |
| 1   | $x_3$ |       | $-3x_1$ | $+x_3$ | $+2x_4$ | $= 10$ |
| 2   | $x_2$ |       | $-2x_1$ | $+x_2$ | $+x_4$  | $= 3$  |



- **Entering Variable:**  $x_1$  (most negative coefficient =  $-7$ )
- **Leaving Variable:** NONE! LP is unbounded.
- How to find the direction of unboundedness from the tableau?

## Example 5: Follow-up

| Row | BV    |       |         |        |         | RHS    |
|-----|-------|-------|---------|--------|---------|--------|
| 0   | $x_0$ | $x_0$ | $-7x_1$ |        | $+3x_4$ | $= 9$  |
| 1   | $x_3$ |       | $-3x_1$ | $+x_3$ | $+2x_4$ | $= 10$ |
| 2   | $x_2$ |       | $-2x_1$ | $+x_2$ | $+x_4$  | $= 3$  |

- Suppose  $x_1$  'enters' the basis and its value is **increased by 1 unit**. Then,
  - due to row-1,  $x_3$  is **increased by 3 units**,
  - due to row-2,  $x_2$  is **increased by 2 units**,
  - $x_4$  is **unaffected** since it is nonbasic.

- $\Rightarrow$  Direction of unboundedness (standard form):  $\begin{bmatrix} d_1 \\ d_2 \\ d_3 \\ d_4 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 3 \\ 0 \end{bmatrix}$
- $\Rightarrow$  Direction of unboundedness (original variables):  $\begin{bmatrix} d_1 \end{bmatrix} = \begin{bmatrix} 1 \end{bmatrix}$

For a **minimization** LP there are two options:

1. **Maximize the negative** of the original objective – no change to the algorithm is needed.
2. Modify the simplex algorithm directly:
  - **Step 3:** if all row-0 coefficients are **nonpositive**, STOP: the current BFS is optimal.
  - **Step 4:** otherwise, choose the variable with the **most positive** row-0 coefficient as the entering variable.

Use the Python file **SimplexTableauWithBV.py** (uploaded to SUCourse) to solve Examples 1–5 with the tableau implementation of the simplex method.